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META-ANALYSIS

Performance of the Asia–Pacific Colorectal Screening score in stratifying the risk of advanced colorectal neoplasia: A meta-analysis and systematic review

Mai Ngoc Luu,[*,†] Nhi Ai Trinh,[‡] Truc Le Thanh Tran,[§] Thinh Phuong Dang,[¶] Toru Hiyama[**] and Duc Trong Quach[*,†]

*Department of Internal Medicine, University of Medicine and Pharmacy at Ho Chi Minh City, Departments of [†]Gastroenterology, [¶]Endoscopy, Nhan Dan Gia Dinh Hospital, [§]Department of Endoscopy, University Medical Center, Ho Chi Minh City, [‡]Department of Internal Medicine, School of Medicine, Tan Tao University, Long An, Vietnam; **Health Service Center, Hiroshima University, Higashihiroshima, Japan

Key words
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Correspondence
Duc Trong Quach, Department of Internal Medicine, University of Medicine and Pharmacy at Ho Chi Minh City, Ho Chi Minh City 700000, Vietnam.
Email: drquachtd@gmail.com

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Abstract

Background and Aim: This study systematically reviewed and meta-analyzed the performance of the Asia–Pacific Colorectal Screening (APCS) score and its incorporation with the fecal immunochemical test (FIT) in stratifying the risk of advanced colorectal neoplasia (ACN).

Methods: We systematically searched for relevant articles in 12 electronic databases and registers on October 20, 2021, and updated the search to September 1, 2023. Random-effect models were used to obtain the pooled performance statistics of the APCS score for ACN risk.

Results: From the 101 records screened, 13 eligible studies in the Asia–Pacific region involving 69 762 subjects who had undergone colonoscopy were enrolled. The pooled prevalences of ACN in the average-risk (AR) tier (APCS 0–1), moderate-risk (MR) tier (APCS 2–3), and high-risk (HR) tier (APCS ≥ 4) groups were 0.9%, 3.1%, and 8.1%, respectively. Compared with the combined AR–MR group, the HR group was significantly associated with a higher ACN risk (pooled diagnostic odds ratio: 2.84, 95% confidence interval [CI]: 2.35–3.45, $P < 0.001$). The APCS score showed a sensitivity of 0.42 (95% CI: 0.40–0.44) and a specificity of 0.86 (95% CI: 0.85–0.86) for predicting the ACN risk, with a weighted area under the curve of 0.642 (95% CI: 0.610–0.657). The combination of the APCS score and FIT substantially improved ACN risk identification, demonstrating pooled diagnostic odds ratios of 4.02 (95% CI: 2.50–6.49) in the AR–MR groups and 5.44 (95% CI: 1.89–15.63) in the MR–HR groups.

Conclusions: The APCS score could effectively stratify the ACN risk in the Asia–Pacific population. Incorporating FIT further improves its performance in identifying high-risk subjects who should be prioritized for colonoscopy screenings.

Introduction

Colorectal cancer (CRC) is one of the most prevalent and deadly malignancies worldwide. As most cases of CRC arise from adenomatous polyps, early detection and endoscopic removal of these polyps have been shown to efficiently reduce both the incidence and mortality rates of CRC.[1–3] Among adenomatous polyps, advanced adenomas, characterized by a diameter of ≥ 10 mm, villous histology, or high-grade dysplasia, progress to CRC at a more accelerated rate, underscoring the importance of their early detection.[4] In recent years, advancements in endoscopic polypectomy techniques, particularly endoscopic submucosal dissection, have offered substantial benefits compared with surgical interventions. Therefore, early detection of advanced colorectal neoplasia (ACN), including high-risk adenomas and CRC, becomes critically important for effective and less invasive treatment.

Current guidelines recommend CRC screening for individuals aged 50 and above and those in high-risk groups.[5–7] Among the various methods for CRC screening, fecal immunochemical test (FIT) and colonoscopy are the main recommended methods in these guidelines.[5–8] However, accessibility to colonoscopy is often constrained by substantial costs and limited resources in several regions. Therefore, using clinical tools that could help stratify the risk levels of having ACN and identify high-risk individuals who should be prioritized for screening colonoscopy is a more cost-effective screening strategy. In this context, the Asia–Pacific Colorectal Screening (APCS) score, a simple tool based on four demographic risk factors (age, sex, family history of CRC, and

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smoking), has been validated and proven to be valuable in stratifying ACN risk across various populations.[9–11] The APCS score, ranging from 0 to 7 points, is calculated based on individual risk factors as follows: age (< 50 years: 0 points; 50–69 years: 2 points; ≥ 70 years: 3 points), gender (female: 0 points; male: 1 point), family history of CRC in a first-degree relative (absent: 0 points; present: 2 points), and smoking history (never: 0 points; current or past smoker: 1 point). This score stratifies patients into three risk tiers: score 0 to 1, average risk (AR); score 2 to 3, moderate risk (MR); and score 4 to 7, high risk (HR).

To the best of our knowledge, no published meta-analysis has comprehensively evaluated the performance of the APCS score. Moreover, several studies suggested that FIT results could be incorporated into the APCS score to help optimize the prediction accuracy for ACN risk.[12,13] Our study aims to systemically review and meta-analyze the performance of the APCS score and the incorporation of FIT into the predictors of this scoring system to predict the risk of ACN.

Methods

Our present systematic review/meta-analysis was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.[14] The study protocol was registered at PROSPERO with the registration number CRD42021288495.

Search strategy. On October 20, 2021, a literature search was conducted in 12 electronic databases and registers, including PubMed, Scopus, Web of Science (ISI), Google Scholar, Cochrane, WHO Global Health Library (GHL), Clinicaltrials, Embase, Graylit Report database (NYAM), SIGLE, Controlled Trials (mRCT), and Virtual Health Library (VHL), to collect all articles reporting the performance of the APCS score to stratify the risk of ACN. We used the following key search: "Asia Pacific Colorectal Screening" OR "APCS score" OR "APCS scores." Following the initial search, an updated search was conducted to include newly published articles up until September 1, 2023.

Inclusion and exclusion criteria. We included any original study that reported the performance of APCS scores in stratifying ACN risk among asymptomatic individuals who had undergone colonoscopy as a screening procedure. These studies were selected without restriction on age, sex, ethnicity, race, language, country, or publication. Asymptomatic individuals were defined as those who did not report experiencing symptoms associated with lower gastrointestinal tract diseases, including rectal bleeding more than once in the past 6 months, unexplained anemia or weight loss, significant alterations in their bowel habits, or lower abdominal pain that typically necessitates medical evaluation.[15] The criteria for asymptomatic screening also included individuals with mild, self-limiting symptoms that had resolved.[10]

Studies were excluded if they (i) were nonoriginal studies or secondary research; (ii) did not have available full texts or abstracts only; or (iii) had nonextractable or unreliable data or overlapped datasets.

Quality assessment. Three independent reviewers assessed the quality of each included article using the Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2) tool.[16] This tool was used to evaluate the study quality of the included studies in four areas, including patient selection, index test, reference standard, and flow and timing. Each study was scored as showing a low, unclear, or high risk of bias for each domain and a low, unclear, or high concern of applicability for all domains except flow and timing. Any disagreements were considered and resolved through consensus.

Data extraction. Studies from the electronic databases underwent title and abstract screening for eligibility by using Covidence, a web-based tool developed for systematic review (https://www.covidence.org/). The full-text screening of the selected articles was performed if they met our criteria for inclusion using the same tool. The standardized extraction sheet was generated after considering data from random articles. This sheet contained basic reference information (title, authors, country, publication year, and journal), study characteristics (study design, study population, and sample size), clinical outcome (ACN), and the predictive accuracy of the APCS score for this outcome (sensitivity, specificity, positive likelihood ratio [positive LR], negative likelihood ratio [negative LR], diagnostic odds ratio [DOR], area under the receiver operating characteristic curves, and 95% confidence intervals [CIs]). For the studies in which data could not be extracted even after contacting their authors by email, those studies were listed as "no data available." Each step was performed independently by three reviewers. Any discrepancies were resolved by discussion to reach a consensus among reviewers and two senior authors.

Statistical analysis. Statistical heterogeneity among studies was evaluated using inconsistency index (I^2) statistics. A fixed-effect model was used if I^2 was < 50%; otherwise, a random-effect model (Mantel–Haenszel method) was applied for the meta-analysis. The pooled sensitivity, specificity, positive LR, negative LR, and DOR were calculated by the DerSimonian and Laird random-effect model. Publication bias was examined using funnel plots. Analyses were conducted using Comprehensive Meta-Analysis software version 3.3.070 and Meta-Disc software (version 1.4, Madrid, Spain).

Results

Literature search. Figure 1 illustrates the flow of the literature selection process. The initial search identified 272 articles, of which 178 duplicate records were excluded. Title and abstract screening was performed for the remaining 94 unique articles. Of these, 62 were excluded because they did not meet our inclusion criteria. We then sought to retrieve the full text of the remaining 32 articles, of which 22 were further excluded for the following reasons: absence of available full text (n = 7), wrong population (n = 3), insufficient data for evaluating the diagnostic performance of the APCS score (n = 11), and data overlap (n = 1). A total of 10 eligible studies were identified in this initial search. An updated search confirmed an additional three eligible studies, resulting in a final sample of 13 studies included in the meta-analysis.[9–13,17–27]

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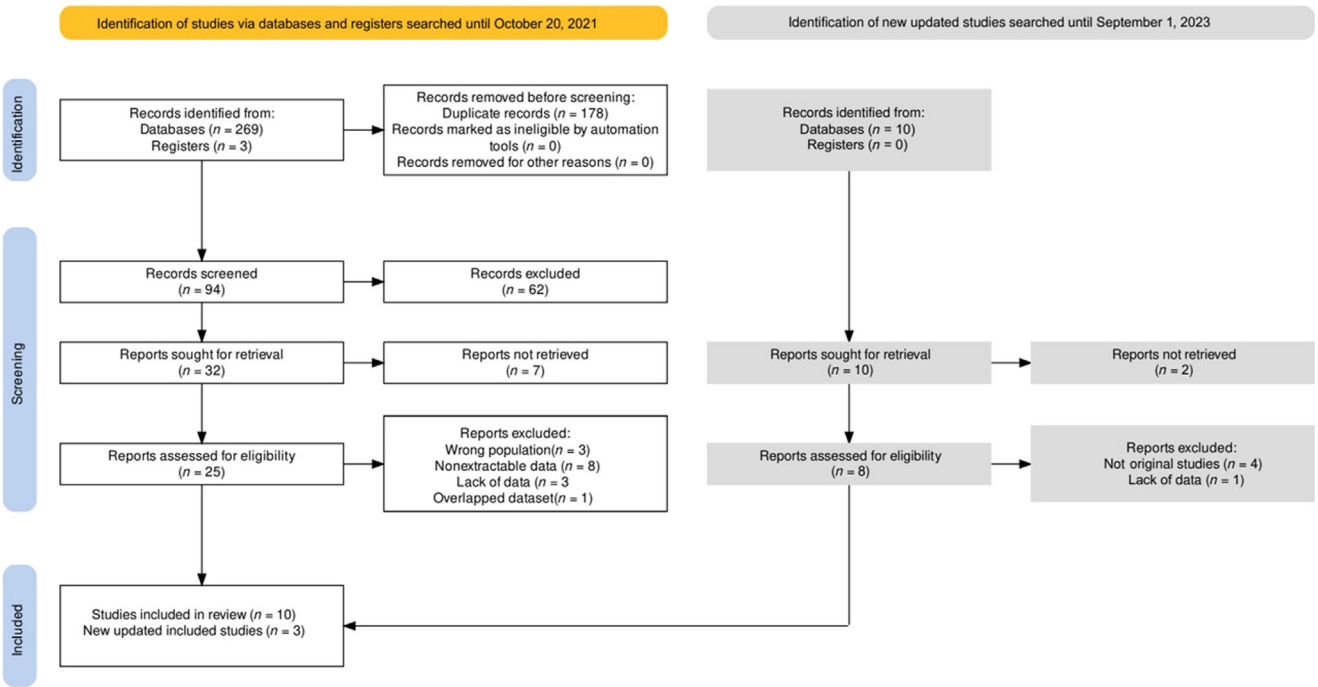


Figure 1 PRISMA flow diagram of the study selection process.

Table 1 Characteristics of studies included in the meta-analysis

Author/publication year/country	Population	*N*	ACN (%)	Mean age (years) (SD)	Male (%)
Yeoh/2011/Asia	Asymptomatic subjects > 16 years	1892	57 (3.0)	51 (11.2)	54
Chiu/2016/Asia–Pacific	Asymptomatic subjects > 40 years	5657	367 (6.5)	57.8 (9)	50.9
Li/2016/China	Asymptomatic subjects 40–75 years	1010	41 (4.1)	52.5 (1.9)	55.1
Jiang/2020/China	Asymptomatic subjects ≥ 40 years	329	30 (9.1)	52.3 (4.8)	43.5
Kong/2022/China	Asymptomatic subjects 50–74 years	1804	112 (6.2)	60.0 (6.1)	50.4
Xu/2022/China	CRC screening subjects 40–85 years	2240	344 (15.4)	60.0 (6.5)	48.1
Zhang/2023/China	Asymptomatic patients > 40 years	812	42 (5.2)	NA	62.7
Kim/2015/Korea	Asymptomatic subjects 30–75 years	1316	56 (4.3)	49.6 (9.9)	42.3
Jung/2017/Korea	Asymptomatic subjects ≥ 30 years	23 023	398 (1.7)	41.4 (8)	73.6
Kim/2019/Korea	Asymptomatic subjects < 50 years	17 873	169 (0.9)	38.9 (5.4)	71.2
Luu/2021/Korea	CRC screening subjects ≥ 40 years	12 520	317 (2.5)	50.9 (7.5)	65.1
Aniwan/2015/Thailand	Asymptomatic subjects 50–75 years	948	113 (11.9)	60.6 (6.7)	35.0
Jangsirikul/2019/Thailand	Asymptomatic subjects 50–75 years	338	25 (7.4)	62.7 (8.6)	38.2

ACN, advanced colorectal neoplasia; CRC, colorectal cancer; NA, not applicable; SD, standard deviation.

Study characteristics and quality assessment. Tables 1 and 2 summarize the characteristics and quality evaluation of the 13 included studies, all of which were conducted in the Asia–Pacific region. No eligible studies were conducted in other regions. Out of the 13 eligible studies, two were multicenter studies: one was conducted in 11 cities in Asia, and another was conducted in 12 countries in the Asia–Pacific area. The remaining studies were single center, with five conducted in China, four in Korea, and two in Thailand. In total, 69 762 subjects, with 2071 patients (3.0%) with ACN, were included in our systematic review and meta-analysis. The criteria for defining ACN were remarkably consistent across the 13 included studies. ACN was generally defined as CRC or an advanced adenoma (at least 10 mm in diameter, at least 25% villous features, or high-grade dysplasia). In addition, the quality assessment showed that most of the assessed studies had a low risk of bias, thus promising good reliability in the findings.

The performance of the Asia–Pacific Colorectal Screening score in stratifying advanced colorectal neoplasia risk. Overall, 39.7%, 45.1%, and 15.2% of participants were stratified as AR tier (APCS 0–1), MR tier (APCS 2–3), and HR tier (APCS ≥ 4). The pooled prevalences of ACN in the AR, MR, and HR groups were 0.9%, 3.1%, and 8.1%, respectively.

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Table 2 Assessment of each study's risk of bias and application concerns utilizing the QUADAS-2 tool

Study	Risk of bias				Applicability concerns		
	Patient selection	Index test	Reference standard	Flow and timing	Patient selection	Index test	Reference standard
Yeoh/2011/Asia	☺	☺	☺	☺	☺	☺	☺
Chiu/2016/Asia–Pacific	?	☺	☺	☺	☺	☺	☺
Li/2016/China	?	☺	☺	☺	☺	☺	☺
Jiang/2020/China	?	☺	?	☺	?	☺	?
Kong/2022/China	☺	☺	?	☺	☺	☺	?
Xu/2022/China	?	☺	?	☺	?	☺	?
Zhang/2023/China	?	☺	☺	☺	☺	☺	☺
Kim/2015/Korea	☺	☺	☺	☺	☺	☺	☺
Jung/2017/Korea	☺	☺	☺	☺	☺	☺	☺
Kim/2019/Korea	☺	☺	☺	☺	☺	☺	☺
Luu/2021/Korea	☺	☺	☺	☺	☺	☺	☺
Aniwan/2018/Thailand	?	☺	☺	☺	☺	☺	☺
Jangsirikul/2019/Thailand	☺	☺	☺	☺	☺	☺	☺

☺, low risk; ?, unclear risk; ☹, high risk.

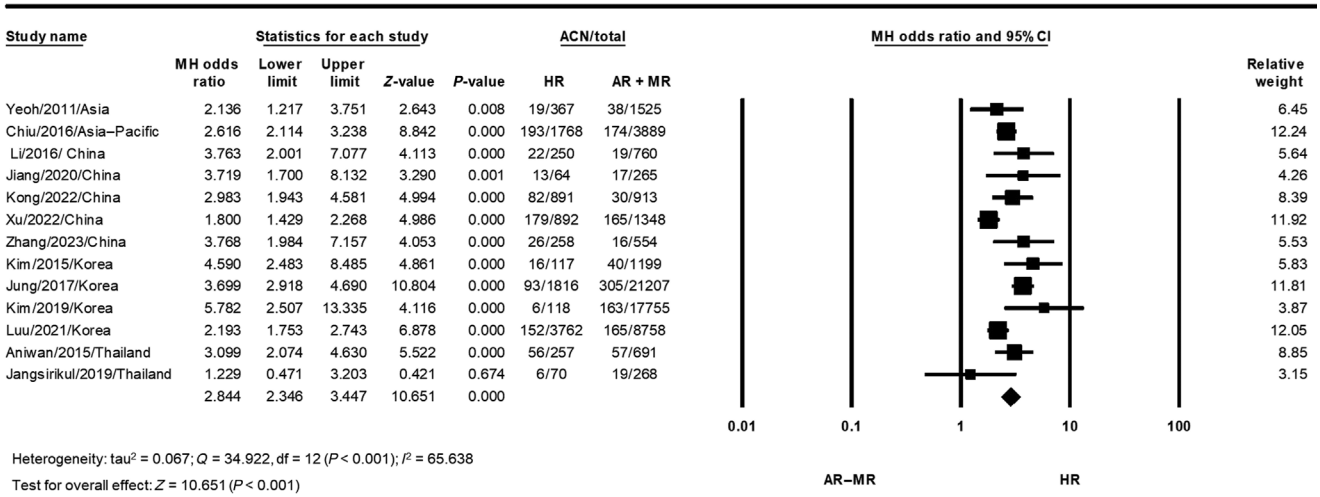


Figure 2 Forest plot demonstrating the association between Asia–Pacific Colorectal Screening score and advanced colorectal neoplasia (ACN) risk. AR, average risk; CI, confidence interval; HR, high risk; MH, Mantel–Haenszel; MR, moderate risk.

The heterogeneity of DOR was shown (Cochran's $Q = 34.92$, $P < 0.001$, $I^2 = 65.64\%$), and the random-effect model was used for meta-analysis. Compared with the combined AR–MR group, the HR group was significantly associated with a higher risk of ACN (pooled DOR: 2.84, 95% CI: 2.35–3.45, $P < 0.001$) (Fig. 2). The pooled sensitivity, specificity, positive LR, and negative LR of the APCS score for predicting the risk of ACN were 0.42 (95% CI: 0.40–0.44), 0.86 (95% CI: 0.85–0.86), 2.01 (95% CI: 1.72–2.35), and 0.74 (95% CI: 0.63–0.85), respectively (Fig. 3). The weighted area under the curve (AUC) value of the APCS score was 0.642 (95% CI: 0.610–0.657), as shown in Figure 4.

Combination of Asia–Pacific Colorectal Screening score and fecal immunochemical test. Figure 5 demonstrates the combination of the APCS score and the FIT result for ACN risk stratification. Three studies investigated this combined approach in participants with an APCS score of ≤ 3 (the AR and MR tiers), revealing a significant association between a positive FIT result and a higher ACN risk (pooled DOR: 4.02, 95% CI: 2.50–6.49, $P < 0.001$). Similarly, two studies examined participants with an APCS score ≥ 2 (the MR and HR tiers), indicating a significantly higher ACN risk among individuals with a positive FIT result compared with those with a negative FIT result (pooled DOR: 5.44, 95% CI: 1.89–15.63, $P < 0.001$).

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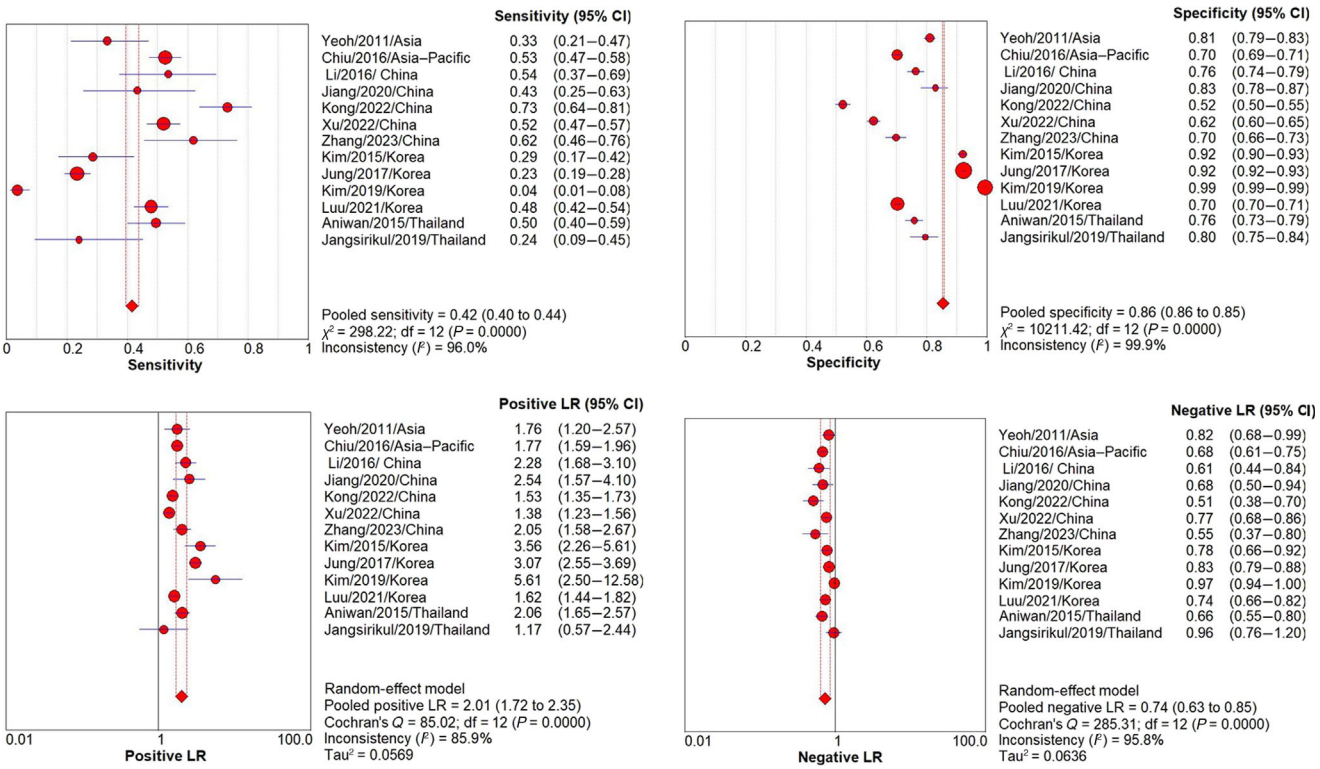


Figure 3 Forest plots demonstrating the performance of the Asia–Pacific Colorectal Screening score for predicting advanced colorectal neoplasia risk. CI, confidence interval; LR, likelihood ratio.

Discussion

The high incidence and mortality rates of CRC emphasize the need for screening strategies to identify and stratify individuals at high risk of developing ACN. In this context, the APCS score, based on a set of demographic risk factors, is a simple and easy-to-apply tool that facilitates the risk stratification of ACN in different population groups. Our systematic review summarizes the available evidence on the APCS score for predicting ACN among asymptomatic populations or CRC screening subjects. A total of 13 studies reporting the performance of the APCS score were identified. Notably, all the studies included in this systematic review were conducted in countries within the Asia–Pacific region. This aligns with the initial development of the APCS score, which was based on the ACN risk factors identified among racial groups in the Asia–Pacific area.[9] This approach raises the possibility of developing similar risk stratification scores for other geographic areas adapted to their specific ACN risk factors. To date, no studies have been conducted to examine the performance of the APCS score in other populations, making its validity somewhat ambiguous in regions outside of the Asia–Pacific region. However, the direct application of the APCS score in non-Asia–Pacific regions may be inappropriate without further validation, given that each region may have its own distinct risk factors and prevalence rates of ACN.

The findings from this meta-analysis reinforce the value of the APCS score in ACN risk stratification, showing a nearly threefold higher risk of having ACN in the HR group than in the AR–MR group. Individuals who fall into the HR tier, characterized by an APCS score ≥ 4, present with at least two of the following risk factors: age ≥ 50, male gender, former or current smoker, or having a family history of CRC.[9] These factors have been shown to be independent risk factors for CRC.[28,29] The presence of at least two risk factors partly explains the higher frequency of ACN lesions in this group compared with the group with a lower APCS score. Based on these results, a colonoscopy should be prioritized for individuals in the HR tier. In addition, the meta-analysis showed a high pooled specificity of the APCS score, indicating its capacity to accurately identify individuals without ACN. This high pooled specificity suggests that the APCS score could reliably reduce unnecessary colonoscopies and effectively serve as a primary screening tool in outpatient clinics, particularly in regions or countries where an organized population-based CRC screening program is not yet in place.

Despite its high specificity, the pooled sensitivity and AUC value of the APCS score in predicting the risk of ACN were found to be only 0.42 and 0.642, respectively. These findings indicate the necessity for further enhancing the efficacy of the APCS score, perhaps through incorporating additional predictive factors or adjusting the current parameters. Our meta-analysis showed that when using only the APCS score to divide patients into three risk groups, the pooled prevalences of ACN in the AR, MR, and HR groups were 0.9%, 3.1%, and 8.1%, respectively. This finding implies that a remarkable portion, approximately 92%, of subjects classified as high risk based on the APCS score were not found

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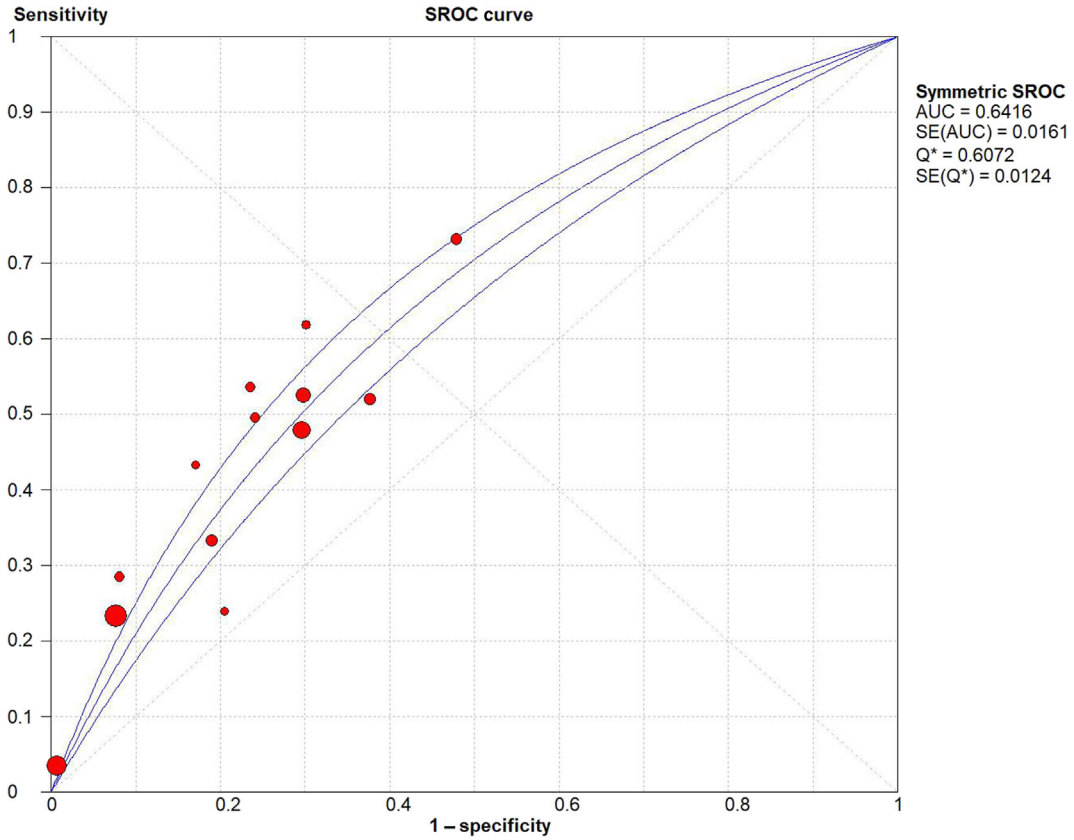


Figure 4 Summary receiver operating characteristic curve (SROC) of the Asia–Pacific Colorectal Screening score for predicting advanced colorectal neoplasia risk. AUC, area under the curve.

to have ACN. Given the significant overload faced by the healthcare system in many countries, particularly in Asian countries, offering screening colonoscopies to all high-risk APCS cases may not be practically implementable. To address this limitation, there have been several studies evaluating the combination of the APCS score with other parameters to improve the capacity of identifying those individuals with ACN and reduce the workload of unnecessary screening colonoscopy. One of the predictors most studied in collaboration with APCS to date is FIT. However, only one published study specifically evaluated the performance of FIT within the HR group, revealing a significant ACN detection rate: 36.9% in the group categorized as HR with a positive FIT result compared with only 11.6% in the HR group with a negative FIT result.[25] Using FIT to further stratify the HR group also demonstrated superior sensitivity and specificity in detecting both CRC and ACN compared with the other groups.[25] Furthermore, the AUC values for the combination of the APCS score and FIT for detecting CRC and ACN were 0.85 and 0.67, respectively. These results suggest that a two-step approach strategy, stratifying with the APCS score and then using FIT to select subjects requiring colonoscopy, is a reasonable choice in areas with limited endoscopy resources.

In addition, despite the relatively low ACN detection rate among individuals classified as the AR and MR tiers based on the APCS score, choosing not to conduct screening colonoscopies on these individuals means accepting the potential risk of either missing or delaying the identification and treatment of ACN cases within these groups. Our meta-analysis demonstrates that the integration of FIT results with the APCS score has shown a substantial improvement in identifying individuals with a higher risk of ACN, even in groups with an APCS score considered to be at low or moderate risk (APCS $\leq$ 3). Notably, the combination approach leads to a significant increase in the ACN detection rate, with a pooled DOR of 4.02, which suggests that a positive FIT result could be a strong indicator of ACN risk among individuals initially categorized as low to moderate risk based on the APCS score. Similarly, an enhanced ACN detection rate was observed in the MR–HR tiers (APCS score $\geq$ 2), where a positive FIT result correlated with a notably higher ACN risk, manifesting a pooled DOR of 5.44. This implies that incorporating FIT results could be a more appropriate risk stratification process, reducing the likelihood of overlooking asymptomatic ACN cases in the lower-risk tiers. However, the potential for missing or delaying the detection of ACN cases in the lower-risk groups remains a considerable concern, necessitating further exploration and optimization of risk stratification strategies.

In the third Asia–Pacific consensus recommendation on CRC screening, statement 2 highlights that multiple risk-stratification systems have been developed and that no single method has proven to be better than the others.[7] It also underscores the utility

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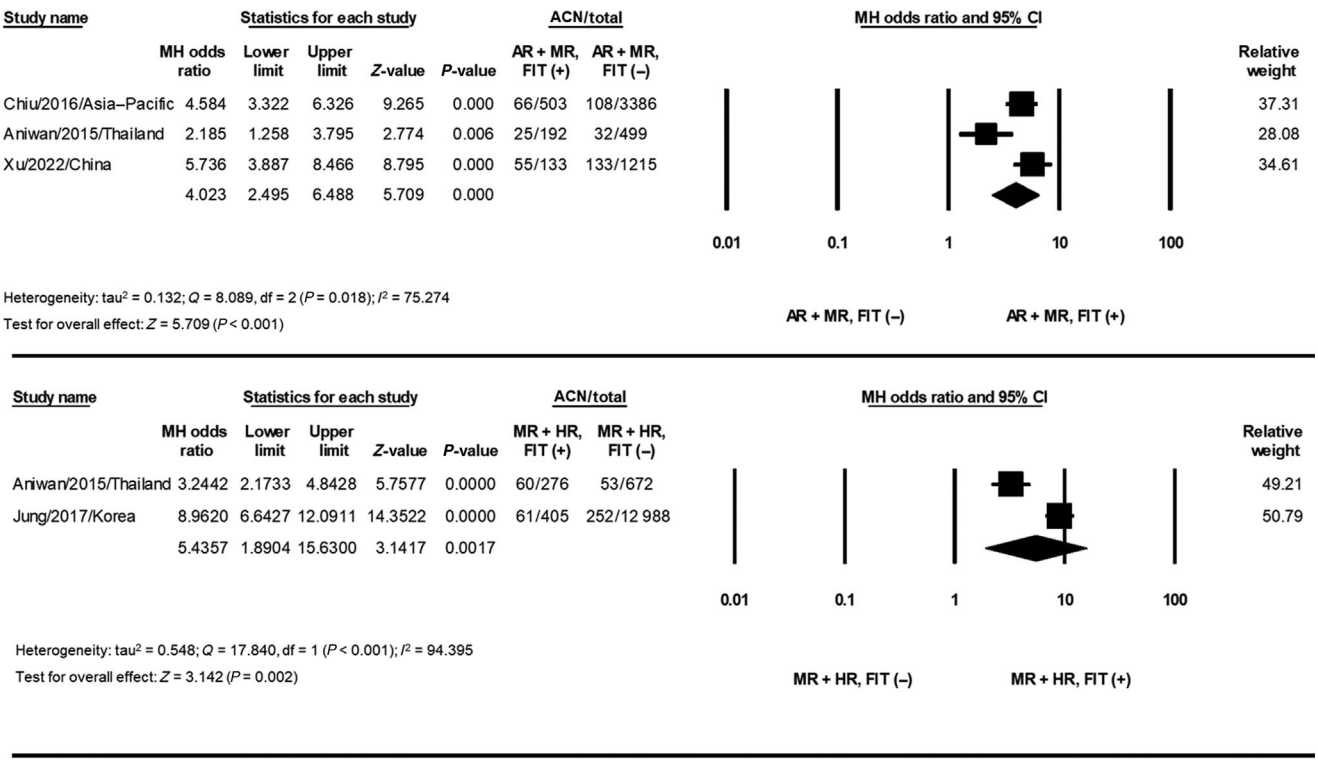


Figure 5 Combination of the Asia–Pacific Colorectal Screening score and fecal immunochemical test (FIT) results in advanced colorectal neoplasia (ACN) risk stratification. AR, average risk; CI, confidence interval; HR, high risk; MH, Mantel–Haenszel; MR, moderate risk.

of the APCS in combination with FIT for efficiently identifying subjects at high risk for ACN. However, the evidence supporting this statement is only at level II-2, derived from the results of well-designed cohort and case–control studies. By systematically reviewing existing studies, our meta-analysis contributes to the body of evidence and could strengthen future recommendations on the utility of this risk stratification approach in the Asia–Pacific region. Although FIT is the primary recommended non-invasive test for CRC screening in resource-limited countries, with a reported sensitivity of 90% and a specificity of 91% for CRC, its sensitivity for detecting advanced adenomas is comparatively low.[30,31] This indicates that FIT might miss a substantial number of these advanced lesions during screening. Our meta-analysis advocates for the integration of FIT and APCS to address this shortfall, particularly for identifying individuals who require a colonoscopy despite testing negative with FIT. Therefore, the APCS score could further enhance the ability to detect ACN if used in combination with FIT in regions or countries where organized FIT screening is already established.

Although our pooled study population extends beyond the recommended age range for CRC screening, which typically spans from 50 to 75 years, it is important to note that the APCS score inherently incorporates age as one of the four demographic risk factors in its calculation, alongside sex, family history of CRC, and smoking. Specifically, individuals below the age of 50 are assigned 0 points in the APCS system. To be classified as high risk by APCS, patients must accumulate a minimum of 4 points, which necessitates being over 50 years old or having other risk factors, such as a family history of CRC. This intrinsic consideration of age within the APCS ensures that individuals within the recommended age range for CRC screening are appropriately assessed and highlights the relevance of APCS in the context of CRC screening guidelines. In addition, recent data indicate a significant increase in the incidence and burden of early-onset CRC in the Asia–Pacific region, with notable rises observed among males in the Western Pacific and females in Southeast Asia.[32] These findings underscore the importance of considering early routine screening for CRC in the Asia–Pacific region. In light of this, the APCS could be applied to the population aged beyond the range recommended in current guidelines.

This study has several limitations. First, there exists potential heterogeneity across the included studies, particularly with regard to the study populations and methodologies, which might have influenced the findings. Second, the APCS score itself does not encompass all possible risk factors for ACN, potentially limiting its predictive accuracy. Third, the limited number of studies analyzing the efficacy of FIT in combination with the APCS score, especially in the high-risk group, also limits the generalizability of the findings.

Conclusions

The APCS score could effectively stratify the risk of ACN in the Asia–Pacific population. Incorporating FIT results further

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optimizes its performance in identifying high-risk individuals who should be prioritized for colonoscopy screenings.

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